# Neural Speech Decoding for Brain Computer Interfaces

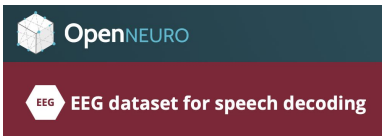
2026.06 – 2026.08. Brain and Cognitive Society 10th

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## Introduction

This project investigates neural signatures associated with inner speech and examines whether phonemic information can be decoded directly from EEG recordings. Using a publicly available speech-related EEG dataset and computational resources, we developed a customized processing pipeline that combines task-specific signal preparation, visual analyses of spatiotemporal EEG dynamics, and a machine learning based phoneme classification framework optimized for decoding performance. Through this work, we sought to assess how discriminative phonetic information is represented in non-invasive electrophysiological signals and to establish a practical foundation for EEG-driven speech decoding brain computer interfaces (BCIs) capable of translating covert speech-related activity into linguistic information.

## 1. EEG Dataset & Experimental Paradigm



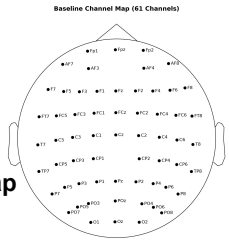
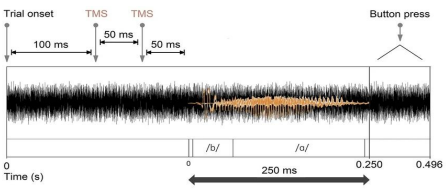




**Figure 1A. Channel Map**

### Dataset & Experimental Paradigm

- 2021 cohort: 16 participants, 64-channel EEG, sampled at 2,000 Hz
- Single phoneme: listen & repeat
- Phoneme pairs (CV): identify the consonant
- Words (CVC): classify real vs. pseudoword
- TMS: paired pulses (50-ms interval), followed by auditory stimulus after 50 ms




(Moreira et al., 2025)

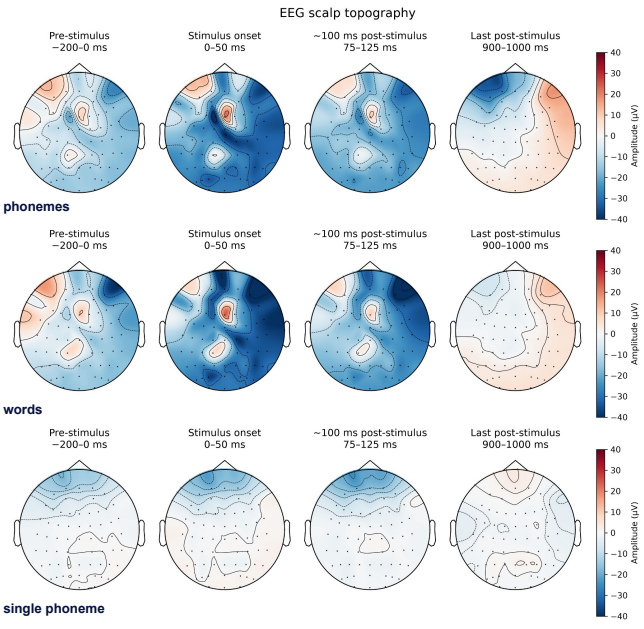



**Figure 1B. Task-Specific Spatiotemporal EEG Topography**

## 2. EEG Preprocessing

### Objective

To remove noise and artifacts from raw EEGLAB recordings and transform the EEG signals into a standardized format suitable for machine-learning analysis.

### Preprocessing Output

The cleaned EEG data were segmented into 1-s epochs and organized into a model-ready 3D tensor:

X = [Trials × 61 Channels × 256 Samples]

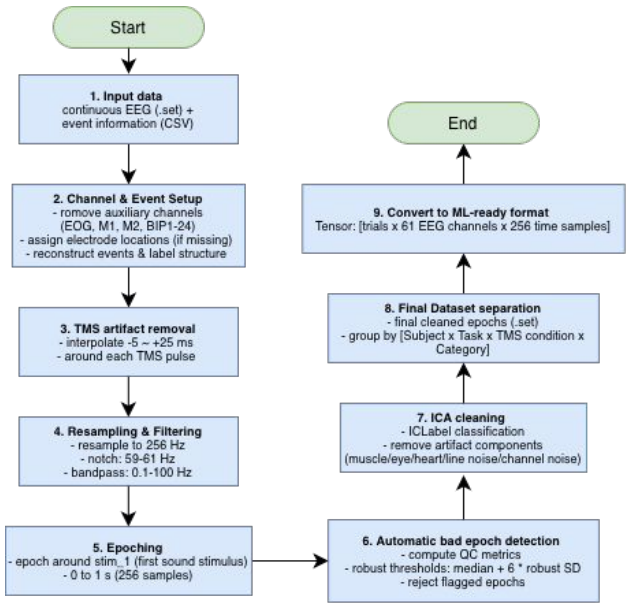



**Figure 2A. Preprocessing Flowchart**

### Data Quality Validation

Epoch-level QC: Robust signal-quality metrics were used to detect and reject contaminated trials automatically.

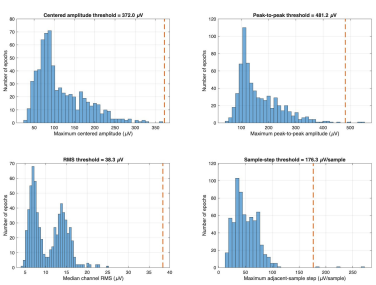


**Figure 2B. Automatic Bad-Epoch Rejection Thresholds**

ICA-level QC: ICLabel-based classification was used to identify and remove residual non-neural components.

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## 3. Stimulus-Locked EEG Signal Characterization

### Data

- Dataset: S02_3_filtered_data.set/.fdt
- Event reference: stim_1 = 0 s

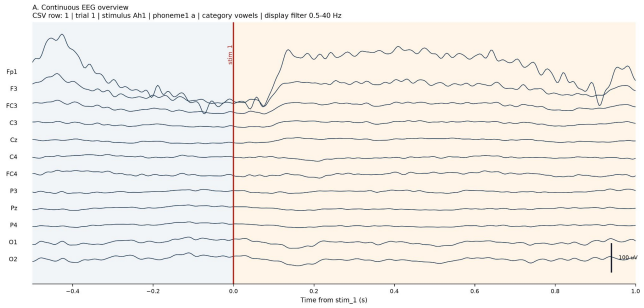



**Figure 3A. Continuous EEG & Event Alignment**

- Single-trial EEG aligned to stim_1 (t = 0 s).
- Confirms temporal synchronization between continuous EEG and auditory stimulus onset

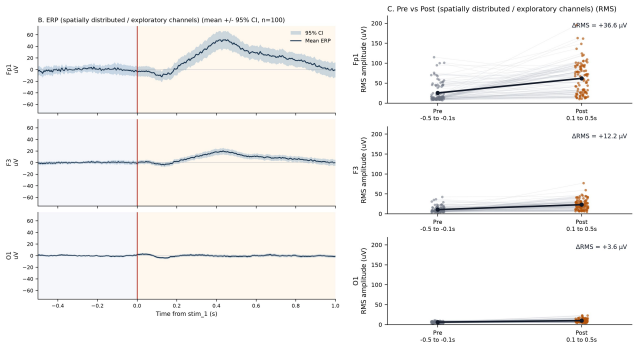



**Figure 3B. Averaged ERP**

- Trials were aligned to stim_1 and averaged.
- A small N100-like negative deflection was followed by a late positive response, peaking at approximately 400–500 ms.
- The response was stronger over frontal/fronto-central electrodes and weaker at O1

**Figure 3C. Pre–Post Stimulus RMS**

- RMS compared between:
  - Pre: −0.5 to −0.1 s
  - Post: 0.1 to 0.5 s
- RMS amplitude increased after stimulus onset, particularly at frontal electrodes.

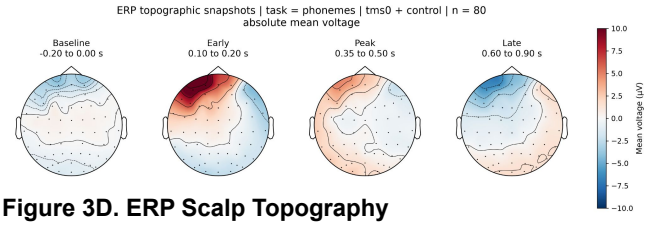



**Figure 3D. ERP Scalp Topography**

- Stimulus-related activity showed a clear time-dependent spatial redistribution.
- Temporal changes across 100–500 ms are consistent with dynamic phonetic-feature processing reported for speech stimuli

## 4. Machine-Learning-Based Model

### Notes

- Our goal was not to build a practically deployable model.
- Rather than prioritizing model speed, our goal was to incorporate into the model architecture the factors that we believe the model should consider, using mathematical formulations and statistical computations.

### Goals

- Subject-specific model: Different LDA for each subject.
- 11-phoneme classification: Classify 11 phonemes (a, i, e, u, o, s, z, p, b, t, d) — chance level: 0.091.
- Input: EEG signals that were recorded while the subject listened to the phonemes.
- Output: One of the 11 phonemes.

### Model

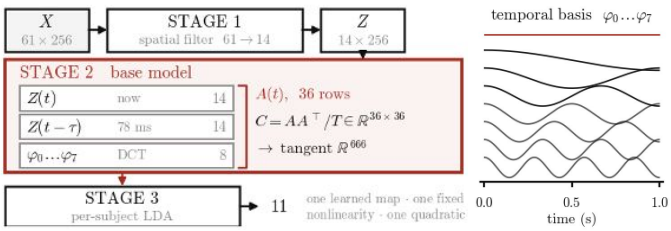



**Figure 4A.** (left) Model Architecture. (right) DCT Temporal Basis.

- **Stage 1**: Noise removal and dimensionality reduction in order to ensure stability
  * $Z = W^T X$
    → Maximize the signal differences between phonemes
- **Stage 2**: Base model; extract information along the temporal and spatial dimensions
  * Delay embedding (τ=20, 78 ms)
    → Capture temporal changes (dynamics)
  * DCT temporal basis ($\varphi_0 \ldots \varphi_7$)
    → Preserve "when" within the representation
  * Riemannian geometry (log-Euclidean)
    → Flatten matrices on a curved manifold into vectors
- **Stage 3**: Extract subject-specific characteristics

### Result

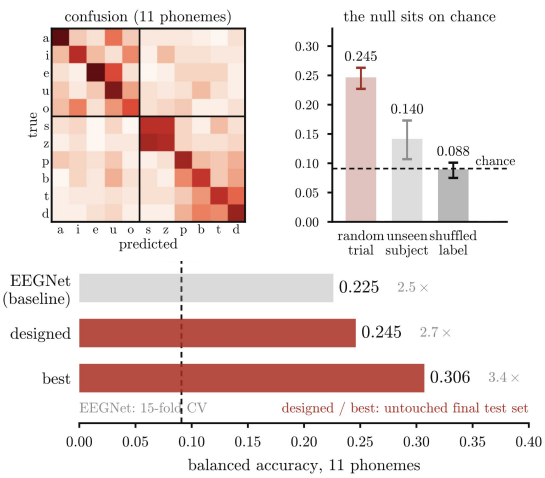



**Figure 4B.** (top left): Confusion Matrix. (top right): Accuracy by Trial.
- Random Trial: Phoneme classification accuracy on trials that were not included in the training data.
- Unseen Subject: Phoneme classification accuracy for a new subject who was not included in the training data.
- Shuffled Label: Classification accuracy when the labels were randomly shuffled.

(bottom): Comparison of 11-phoneme classification accuracy among EEGNet, our model (designed), and the additional experimental model (best).